A BEAUTIFUL NEGATIVE RESULT

WHEN THE MAP IS WRONG, THE BETTER WARNING SIGN LOST.

A story about a robot, a missing wall, and a metric that looked smarter than the evidence allowed.

THE VERDICT

Thesis rejected.

Candidate-Set Regret did not outperform the required collision-risk baselines.

CSR / R0	CSR / R1	CSR / R2
0.504	**−0.470**	**−0.307**

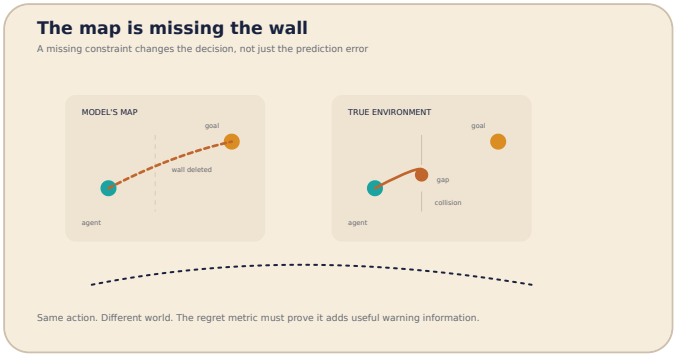
Required N: ∞ for the key comparisons. No confirmatory run.

01 THE SETUP

THE ROBOT AND THE MISSING WALL

Imagine a robot trying to reach a goal. Its mental map is missing a wall that exists in the real world.

We are not trying to prove that a missing wall is dangerous. We already know the model is wrong. The real question is subtler: **can regret tell us where the danger is, better than simpler measurements?**



▶ What is Candidate-Set Regret?

02 THE TARGET

FIRST, WE HAD TO FIX A BROKEN EXPERIMENT.

If every checkpoint is safe, there is no risk to rank. The original R0 design produced exactly that: **768 checkpoints, zero collisions, zero variance.**



AMENDED R0 PASSED EVERY GATE

Variance	**0.163**	≥ 0.02
Interior band	**34.2%**	≥ 20%
Exact endpoints	**58.3%**	≤ 70%
Split-half reliability	**0.9924**	≥ 0.75

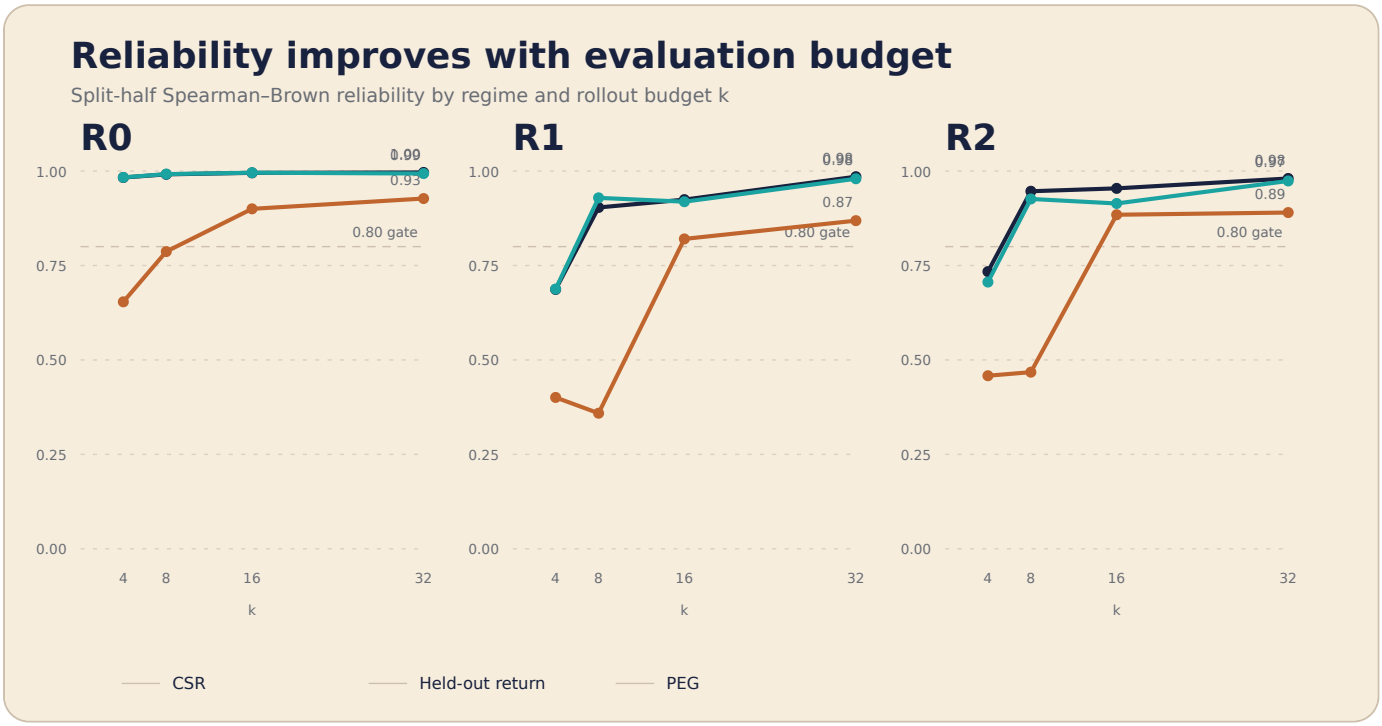
Randomized goals and broader speeds produced safe, risky, and uncertain checkpoints. R1 and R2 also passed with reliability above 0.95.

03

RELIABILITY

WE CHECKED THE MEASUREMENTS BEFORE TRUSTING THEM.

More episodes usually mean more stable estimates, but compute is not free. We swept \(k=\{4,8,16,32\}\) and chose the smallest budget that cleared the reliability gate in every regime.



Selected \(k=16\) smallest budget with CSR and held-out-return reliability above 0.80 in every regime

04

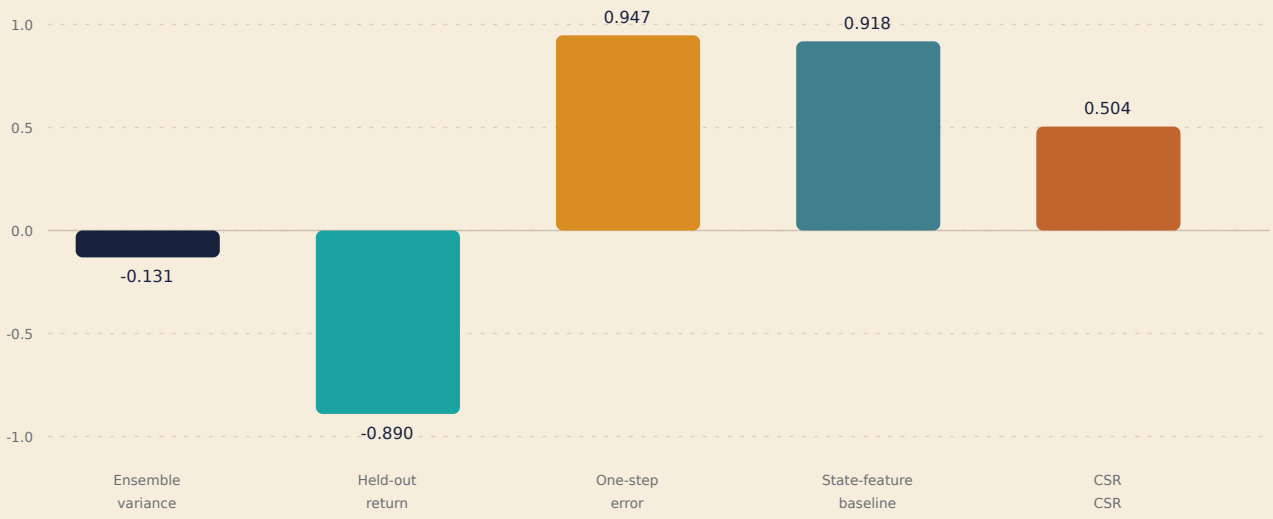
THE TOURNAMENT

THE METRIC THAT LOOKED SMARTER LOST.

Each score ranks checkpoints by predicted collision risk. We compare its Spearman correlation with independently estimated collision probability. **Higher is better.**

R0: risk-ranking comparison

Mean per-seed Spearman correlation with independent collision probability · higher is better



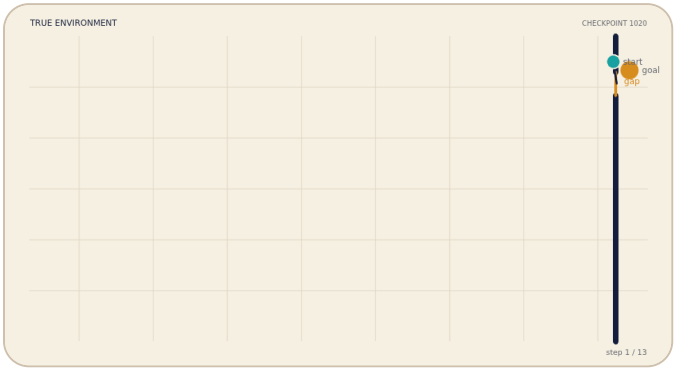
SCORE	SPEARMAN P	REQUIRED N	WITHIN CAP?
Ensemble return variance	-0.131	6	Yes
Held-out return	-0.890	1	Yes
One-step error	0.947	Infinity	No
State-feature baseline	0.918	Infinity	No
Candidate-Set Regret (CSR)	0.504	–	N/A

The state-feature baseline is fitted on development seeds only. It uses geometry, gap, velocity, and noise—not model predictions.

05

THE TRAJECTORY

ONE REAL TRAJECTORY, NOT A STYLIZED SUCCESS.



WHY THIS CHECKPOINT?

Among model-seed-0 R1 checkpoints, choose the collision probability nearest to the median of all 16 R1 model seeds; break ties by lowest checkpoint id.

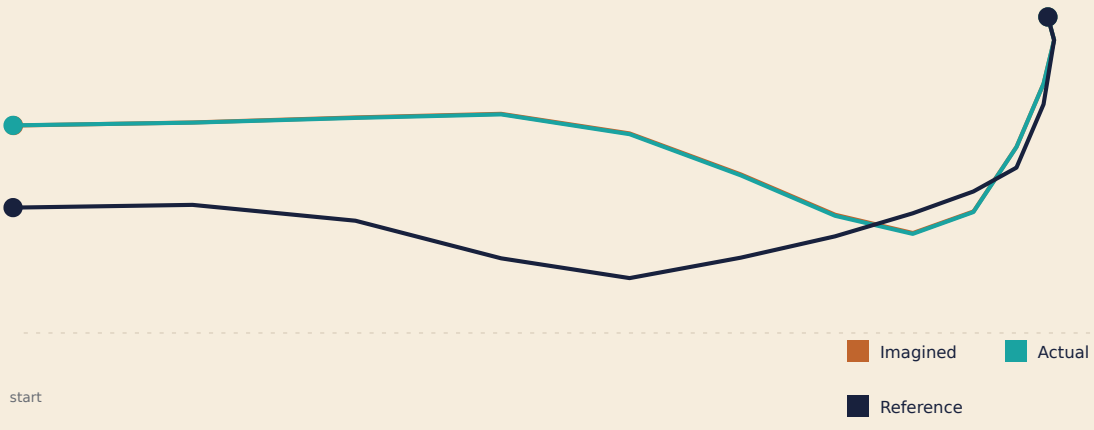
Checkpoint	1020
Median collision p	0.7656
Start state	-0.26, 0.22, 0.92, -0.06
Goal	1.50, 0.00
Gap center	-0.351
Gap width	0.60
Noise σp / σv	0.02 / 0.02

▶ **What do the colors mean?**

▶ **Read the raw trajectory**

A real trajectory, not a stylized success

Checkpoint 1020 · actual model execution in the true environment

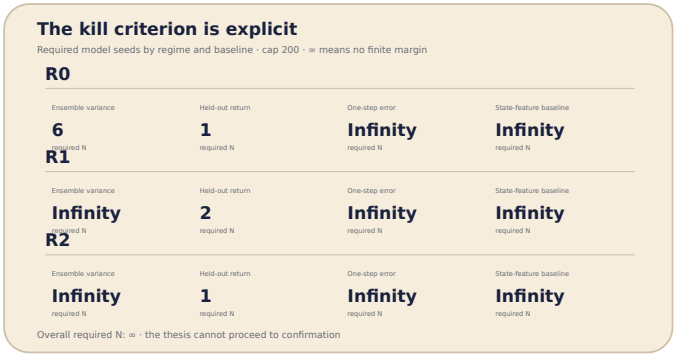


06 THE VERDICT

A USEFUL NEGATIVE.

CSR did not beat every required baseline in even one regime. The required sample size for the key comparisons was unbounded, so the pre-registered kill rule was triggered.

No confirmatory experiment will run. This does not prove CSR is useless everywhere. It shows that in this deleted-wall testbed, geometry and one-step error provide much better collision-risk predictions.



01

NEGATIVE IS INFORMATIVE

A pre-registered rejection is evidence, not a failed project.

02

DO NOT TUNE UNTIL IT WINS

Increasing seeds or redesigning until CSR looks better would undermine the protocol.

03

PRESERVE THE EVIDENCE

Raw JSONL, manifests, hashes, trajectory records, and analysis remain available for review.

PROVENANCE & REPRODUCIBILITY

EVERY NUMBER HAS A SOURCE.

SOURCE ARTIFACT HASHES

development-targets/target_quality.json	f1ba749ff026907f2eeeeef2a1958bb29571315f9127273509a1244bd5df0414
development-targets-r0-broadened/target_quality.json	7c44a2d421f8f12b603724150ab15f0a0e5be11e8453585009de86d2abe04050
development-sweep-r0-broadened/sweep_summary.json	2c076a6605a7c129431e9c97c799d8736a61c4790bc80e37409bfcc552e47fff
development-power-r0-broadened-k16/power_analysis.json	a7c2f3762613edc76e49b9e1e3f89b5737e6827673432dcf802773be63b37317

REPRODUCE THE ANALYSIS

```
PYTHONPATH=src python -m counterworld.pilot --stage power --output artifacts/development-power-r0-broadened-k16 --primary-k 16 --workers 8 --random-goals --r0-speed-range 0.75 1.5
```

Run from the research/ directory with Python 3.11+ and NumPy 2.x. Artifacts remain development-only.